UPVAAS-Twin

A festival-aware metabolic digital twin for Indians with type 2 diabetes

Predicting hypoglycaemia and glucose spikes two hours ahead, and rehearsing a patient's religious fast before the patient lives it.

Happiest Health · Reimagining and Reforming Healthcare in India Summit 2026

Digital Twin Challenge

Breakpoint Consulting · Symbiosis Institute of Management Studies, Pune

Half of Indian T2D patients fast for religion, most take a drug that does not stop

The dangerous combination is scheduled, so it can be predicted

The scale

101 million adults with diabetes in India.

136 million more with prediabetes.

ICMR-INDIAB, Lancet Diab Endo 2023

The drug

57% of one North Indian clinic cohort take a sulfonylurea.

It keeps releasing insulin even when no food arrives.

The behaviour

49% had fasted for religion in the past year.

4% checked their glucose during the fast.

1.35% asked a doctor first.

Navratri · Ekadashi · Karva Chauth (nirjala) · Ramadan roza

A morning glimepiride followed by a thirteen hour fast produces a hypoglycaemia that was predictable the day before. The festive thali that breaks the fast produces a spike that was equally predictable.

Existing digital twins treat those days as noise. They predict meal responses, assume the patient wears a CGM, and never look at the calendar.

What we predict

Adverse event	Threshold	Horizon	Who it matters for
Hypoglycaemia	< 70 mg/dL for 15 min	2 hours ahead	Sulfonylurea patients during a fast
Hyperglycaemia	> 250 mg/dL for 15 min	2 hours ahead	Everyone, after the fast breaks

Commercial twins answer a different question, and assume hardware most Indians lack

Where UPVAAS-Twin sits against what already exists

	Twin Health / January AI	Academic hybrid twins	UPVAAS-Twin
Predicts	food to glucose response	glucose trajectories, mostly T1D	adverse events 2 h ahead, plus what-if
Model	gradient boosting / RF	physics-constrained neural net	mechanistic ODE + particle filter + calibrated GBM
Uncertainty	not published	sometimes, marginal only	conformal, reported by glucose range
Needs a CGM?	yes	yes	no, three sensor tiers evaluated
Validated against	not published	the same model family	a structurally different body model
India specific	food database	no	festival calendar, thin-fat prior, SU pharmacogenomics, ASHA workflow
Fairness audit	no	rarely	yes, led by fasting tradition

A 2026 review of 28 diabetes digital twin studies found one that addressed fairness and three with a human in the loop. None forecast adverse events.

Static data sets the prior, live data updates the posterior, the calendar says what is coming

Five layers, from raw streams to a clinician's screen

1 Data in

India-localised synthetic EHR:
labs, meds, eGFR, CYP2C9, TCF7L2
CGM or 2 fingersticks a day
watch: HR, HRV, steps, sleep
festival calendar + PM2.5

2 Twin core

EHR to physiology prior (ridge)
SU-Bergman-Gut ODE
particle filter, 64 candidate
physiologies per patient
2 h forecast on the calendar plan

3 Intelligence

gradient boosting on fused
EHR + stream + calendar + twin
isotonic calibration
Mondrian conformal bands
counterfactual drivers

4 Decision

clinician console, ranked roster
Vrat Rehearsal what-if
ASHA pre-festival recall list
FHIR R4 RiskAssessment

5 Governance

DPDP: consent ID, purpose limit
ABDM: HIE-CM consent artefact
CDSCO draft: informs a doctor,
likely Class B
ICMR: human in the loop

What makes it a twin rather than a classifier

The particle filter keeps 64 candidate physiologies per patient and re-weights them on every glucose reading, so the model follows the patient instead of scoring them once.

Those same posterior particles are replayed forward under a changed plan. That is the Vrat Rehearsal: the doctor asks what happens if the glimepiride moves to the fast-breaking meal, and gets an answer before the fast.

The model knows why a fast plus a morning glimepiride is dangerous

Mechanistic core, so every alert can be explained to the patient

SU-Bergman-Gut model, per minute

Bergman minimal model for glucose and remote insulin action.

Two-compartment gut, absorption set by the food's glycaemic index and fat.

Sulfonylurea PK/PD that keeps secreting insulin at low glucose.

Fasting clock: hepatic output falls after about 11 hours without food.

Counter-regulation blunted by autonomic neuropathy.

CYP2C9 genotype and eGFR lengthen the drug half-life prior.

Why that combination harms people

The tablet is taken at 08:30 on a normal schedule.

No carbohydrate arrives for the next thirteen hours.

Insulin secretion continues anyway.

After eleven hours the liver has less glycogen to release.

In a patient with neuropathy the warning symptoms are blunted.

Glucose falls in the late afternoon, hours before moonrise.

The twin turns that mechanism into four things a doctor can act on

Risk now

Calibrated probability of an event in the next two hours, with the drivers behind it.

A virtual patient

Personalised insulin sensitivity, beta-cell gain, hepatic set-point and drug half-life, each next to its EHR-only prior.

A rehearsal

Tomorrow's fast replayed under four drug plans before the patient lives it.

A recall list

Every faster in the ward ranked before the festival, with language, ward and an action script.

The twin earns its keep where there is no CGM, which is most of India

200 synthetic patients x 28 days, 5-fold patient-grouped cross-validation

Tier B: two fingersticks a day and a basic smartwatch. No CGM.

0.08 → 0.13

hypoglycaemia AUPRC

1.6x by adding the twin

5.4 → 2.7

false alarms per patient-day

the twin filters as well as ranks

26%

precision of a 20-name worklist

against a 0.9% base rate

Of the 20 patients the twin flags each morning, about 5 really do have an event within two hours.

Sensor tier	AUROC	AUPRC [95% CI]	Without twin	P@20	Recall	False alarms/day
A CGM + watch	0.959	0.220 [0.10-0.34]	0.171	0.36	56%	1.00
B 2 fingersticks + watch	**0.911**	**0.132 [0.05-0.23]**	**0.080**	**0.26**	**58%**	**2.69**
C EHR + phone only	0.792	0.071 [0.02-0.14]	0.034	0.26	62%	6.87

Forecasting, tier A

Twin RMSE 27 / 33 / 38 mg/dL at 30 / 60 / 120 min.

Persistence 29 / 44 / 55 mg/dL.

Clarke error grid zones A+B: 97.2%.

How it was scored

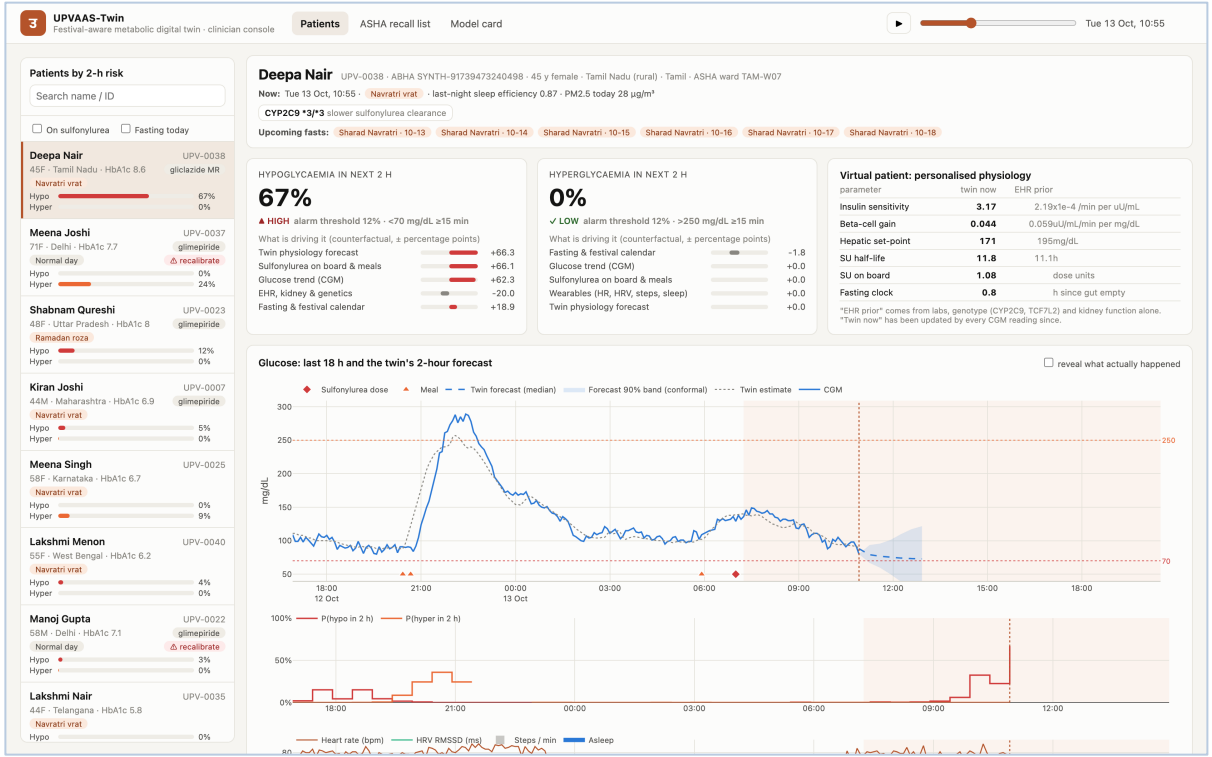
Every patient scored by a model that never saw them, in the dashboard as well as the metrics.

Patient-clustered bootstrap intervals on every headline number.

Only 40 of 200 patients ever had a hypoglycaemic event, so the intervals are wide and we print them.

The doctor sees the risk, the reason, and the patient's own physiology

Clinician console, built for a ten-minute OPD consultation



Roster ranked by two-hour risk, with today's vrat type and a recalibrate flag

Risk, with its drivers

Calibrated probability and the alarm threshold.

Counterfactual attribution shows which feature group moved it.

The virtual patient

Insulin sensitivity, beta-cell gain, hepatic set-point, drug half-life.

Each shown against its EHR-only prior, so the doctor sees what the data changed.

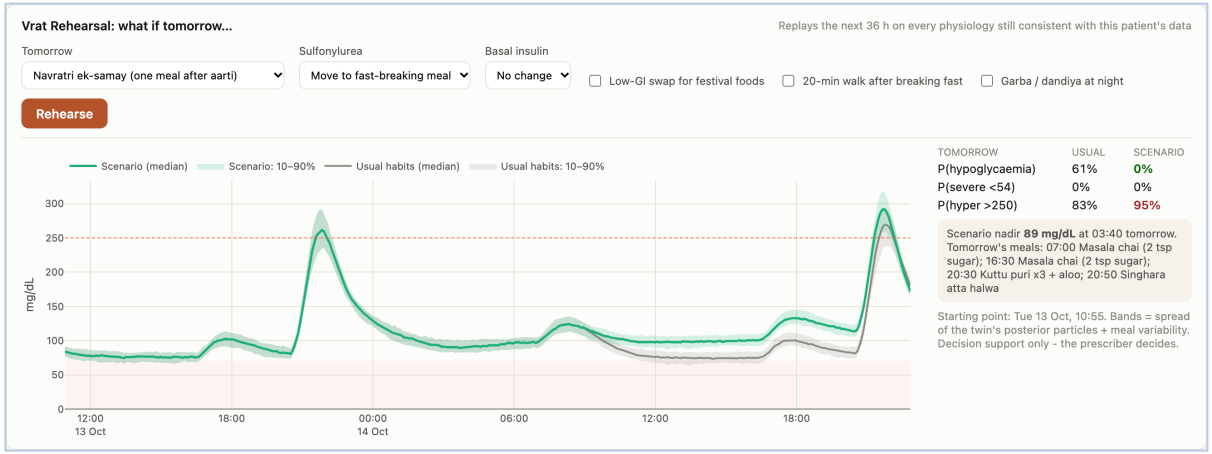
Forecast band

Mondrian conformal, so the band widens where the twin is least reliable.

A reveal toggle shows what actually happened, for teaching.

Moving the tablet to the fast-breaking meal takes this patient from 61% to 0%

Vrat Rehearsal: a 36-hour counterfactual on the posterior particles



Grey is the patient's usual habit, green is the proposed change

What the doctor changes

Which fast tomorrow is: Karva Chauth, roza, Navratri phalahar or ek-samay.

Sulfonylurea: as usual, halved, skipped, or moved to the fast-breaking meal.

Basal insulin, a low-GI swap, a walk after iftar, garba at night.

What comes back

P(hypo), P(severe), P(hyper) and the predicted nadir time.

Across all sulfonylurea-treated fasters, moving the dose lowers predicted hypoglycaemia from 28% to 10%.

Read as a ranking, not a calibrated probability. The rehearsal over-predicts how often a fast ends in hypoglycaemia.

We made the benchmark hard on purpose, and published the baseline that beats us

A result you cannot break is a result you cannot trust

The virtual body is not the twin

The body runs a different model: three-compartment gut, interstitial insulin delay, circadian insulin sensitivity, renal glucose excretion, an incretin term.

The twin has an equation for none of them, so the error we report includes model misspecification and not only parameter error.

Eating is messy on purpose

Correlation between the festival meal plan and what was actually eaten: 0.55.

23% of food intake was never on the plan.

Wide meal-time jitter, skipped meals, unplanned snacks. The calendar is a prior, not an oracle.

The baseline that keeps us honest

Four features: planned carbohydrate in the next two hours, HbA1c, and time of day. No glucose reading, no wearable, no twin. It reaches hyperglycaemia AUPRC 0.30 against 0.60 for the full tier-A model.

We publish it so a reader can see how much of the task is physiology and how much is a meal timetable.

Null results we report rather than bury	Number
The festival calendar as model features adds nothing	0.218 vs 0.220
The drift self-check does not work well enough to deploy	AUROC 0.53
Forecast bands cover far less than 90% above 250 mg/dL	58% vs 90% target
The hypoglycaemia signal rests on a minority of patients	40 of 200

We took it to real patients, it did not beat persistence, and the failure explains itself

ShanghaiT2DM: 107 real CGM recordings, no retraining. Beyond what the brief asks for

2-hour forecast on real patients	Twin	Persistence
RMSE	52 mg/dL	46 mg/dL
Clarke zones A+B	91.4%	95.5%
On our own simulator	38 mg/dL, ahead of persistence	55 mg/dL

Can a better prior fix it? (59 held-out patients)	Ratio to persistence
Original prior, as submitted	1.10
Original prior, simply widened	**1.01**
Prior refitted on real patients	1.10

The finding is worth more than a fixed number

Refitting the prior on real patients changed nothing. Merely widening it reached parity with persistence and no further.

The refit told us why. Its residual spread exploded: hepatic set-point SD went from 8 to 40 mg/dL and ln insulin sensitivity from 0.22 to 1.26. In a real cohort the routine EHR fields barely predict these physiological parameters. In our simulator they predict them well because the simulator generated the labs from those parameters in the first place.

What this changes

Not the design. The mechanistic core, the filter, the rehearsal and the sensor ladder are unchanged.

It puts a number on the warning the README already carried: the simulated results are an upper bound.

What would have to happen next

Fit the prior against real physiological reference data, from clamp or OGTT studies, not routine labs.

Handle sparse 15-minute sampling directly instead of upsampling.

Then repeat this same script and see whether the numbers move.

Audited first on the axis this project is built on

Fairness, uncertainty and the Indian regulatory path

Fasting tradition	AUROC	Events
karva chauth	0.980	76
navratri	0.938	1245
none	0.981	673
ramadan	0.973	404

Uncertainty that keeps its promise

A single conformal margin covers 47% of outcomes above 250 mg/dL while claiming 90%.

Mondrian conformal, calibrated per predicted-glucose band, raises that to 58% with a narrower band overall.

Both are published, by outcome range.

Also audited

Sex, urban and rural, age, eGFR, region.

A model card ships inside the dashboard.

The tier C path still works for a patient who refuses or cannot afford a wearable.

Requirement	How UPVAAS-Twin meets it
DPDP Act 2023 / Rules 2025	Synthetic data only. Production design keeps a consent ID on every record, enforces purpose limitation, deletes on expiry.
ABDM	Registers as a Health Information User. Consent artefact via HIE-CM. FHIR R4 bundles validated against the R4 schema.
CDSCO draft software guidance, Oct 2025	Informs a registered practitioner rather than directing treatment, so likely Class B. Drift monitor is the hook an Algorithm Change Protocol would attach to.
ICMR 2023 AI ethics	Human in the loop, Indian-representative priors, subgroup audit, published model card, a working path for patients without devices.

It runs on a clinic laptop, and reaches the patient through an ASHA

Where this goes next, and what it would take

1. Two weeks before

The ASHA app receives a ranked recall list for every faster in the ward, with language, ward and an action script.

2. Home visit or call

The highest-risk patients are nudged to see a doctor before the fast, not after an ambulance call.

3. Teleconsult

The doctor runs a Vrat Rehearsal in clinic or over eSanjeevani and picks a drug plan.

4. During the fast

The patient has a dose-timing plan in their own language. The twin keeps scoring risk every 30 minutes.

Where it would run

NP-NCD screening at Ayushman Arogya Mandirs, which carry about 25 million people on diabetes treatment under the 75 by 25 initiative.

Private diabetology clinics and CGM programmes.

One clinic server can cover a district. Tiers B and C need no CGM.

How we would prove it

Fit the prior on real physiological reference data, then repeat the external validation.

IRB-approved observational pilot during Navratri 2027, 100 sulfonylurea-treated fasters on Libre CGMs.

Cluster-randomised ASHA recall-list trial, outcomes being hypoglycaemia events and ER visits.

The fast is on the calendar. The hypoglycaemia does not have to be a surprise.

Research prototype on synthetic data. Not a medical device. It informs a registered medical practitioner and decides nothing.

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